Irrigation and the Weather Sensitivity of U.S. Crop Insurance Losses

Bruno Pinto[†] Taro Mieno[‡]

June 2026

[†]Teaching Assistant Professor, Department of Agricultural Economics and Agribusiness, University of Arkansas. Email: brunoc@uark.edu.

[‡]Associate Professor, Department of Agricultural Economics, University of Nebraska–Lincoln. Email: tmieno2@unl.edu.

Abstract

This paper examines how extreme heat and water deficit relate to federal crop insurance losses for corn and soybeans, and how that relationship differs between irrigated and non-irrigated production. Using county-level administrative data from the USDA Risk Management Agency for 2011–2023, restricted to Revenue Protection policies, we estimate how Extreme Degree Days above 29°C and growing-season water deficit relate to two outcomes, indemnity per acre and the loss cost ratio. Each weather variable is interacted with irrigation status. Because irrigation is not randomly assigned, we treat every irrigation differential as a descriptive association rather than the causal effect of applying water. Operations classified as irrigated exhibit a much smaller association between extreme heat and insured losses than non-irrigated operations. For corn the implied heat sensitivity of the loss cost ratio is about 99% smaller under irrigation, and the residual association is statistically indistinguishable from zero; for soybeans it is about 81% smaller. Water deficit follows a U-shaped relationship with losses, with both excess moisture and drought raising indemnities. The two practices diverge under drought-like conditions but respond similarly under excess moisture. These patterns are robust to a smooth temperature response, alternative fixed effects, and a correction for binning bias. Translating the estimates into projected premium rates and federal outlays under uniform warming, the program's projected fiscal exposure is concentrated in non-irrigated production: projected additional annual outlays for corn and soybeans rise to roughly $5.6 billion at +3°C, with non-irrigated production accounting for the overwhelming majority in every scenario. The concentration is descriptive of the existing stock of irrigated acres rather than evidence that irrigation causes lower fiscal exposure.

JEL Codes: Q12, Q14, Q54, G22

Keywords: climate adaptation, crop insurance, extreme heat, federal crop insurance program, irrigation, water deficit

1. Introduction

The Federal Crop Insurance Program (FCIP) is the central pillar of the U.S. farm safety net, with annual indemnities and producer-premium subsidies that together exceed $10 billion in most years and rise sharply in major loss years. As the climate warms, the program's fiscal exposure has become a first-order question for agricultural policy. Diffenbaugh, Davenport, and Burke (2021) attribute approximately $27 billion in cumulative U.S. crop insurance losses over 1991–2017 to historical temperature trends, and find that warming contributed nearly half of the losses in 2012, the program's worst single year.

Irrigation is the on-farm technology associated with reduced yield impacts of extreme heat and atmospheric moisture stress (Carter et al., 2016; Tack, Barkley, and Hendricks, 2017). Most existing evidence on irrigation as a climate adaptation strategy speaks to mean yields and yield distributions. Yet what matters fiscally for the FCIP is not mean yields but indemnities. Under the federal crop insurance rating, the expected loss cost ratio equals the actuarially fair component of the premium rate, and expected indemnity per acre equals the actuarially fair premium per acre (Tack, Coble, and Barnett, 2018). Existing work has not directly quantified how the program's warming exposure splits across the level at which the FCIP actually rates contracts: crop, policy type, and irrigation practice.

The closest antecedent to the projection we develop is Tack, Coble, and Barnett (2018), who show that warming would raise crop insurance premium rates and government outlays, with substantial cross-sectional heterogeneity. Their projections are constructed at the aggregate program level and do not separate production by irrigation practice. We build on that work in two ways. We move from the backward-looking attribution of Diffenbaugh, Davenport, and Burke (2021) to a forward-looking projection under counterfactual warming, and we disaggregate the projection to the crop-policy-practice level so that we can ask where the program's exposure is concentrated rather than only how large it is in aggregate.

We use county-level administrative data from the USDA Risk Management Agency (RMA) for 2011–2023, restricted to Revenue Protection (RP) policies for corn and soybeans, the two largest commodities in the FCIP. We estimate how Extreme Degree Days above 29°C and growing-season water deficit relate to indemnity per acre, in dollars, and to the loss cost ratio. Each weather variable is interacted with irrigation status, so the interaction terms describe the differential weather sensitivity of irrigated relative to non-irrigated operations within the same county and year. Because irrigation status is not randomly assigned, we treat the irrigation interactions throughout as associations rather than as the causal effect of applying irrigation water. We cannot separate the role of supplemental water from selection into irrigation, soils within counties, and management practices that travel with irrigation status, and we are explicit about this limitation wherever we

report a differential. We then translate the estimated coefficients into projected changes in FCIP premium rates and program subsidy outlays under uniform-warming scenarios of $+1$°C, $+2$°C, and $+3$°C, applying the daily-temperature approach of Schlenker and Roberts (2009) to GridMET temperatures (Abatzoglou, 2013).

Three findings emerge. First, operations classified as irrigated exhibit a substantially smaller association between extreme heat and insured losses than non-irrigated operations. For corn, the implied heat sensitivity of the loss cost ratio is approximately 99% smaller under irrigation, and the residual association is statistically indistinguishable from zero. The corresponding figure for soybeans is approximately 81%. Second, water deficit follows a U-shaped relationship with insured losses, with both excess moisture and severe drought associated with higher indemnities and higher loss cost ratios. Irrigated and non-irrigated operations differ substantially under drought-like atmospheric conditions but respond nearly identically under excess moisture, an asymmetry consistent with the biophysical role of supplemental water rather than with general differences in operator quality across practices. Third, projected FCIP fiscal exposure rises sharply with warming for non-irrigated corn and soybeans but barely changes for irrigated operations. At $+3$°C of uniform warming, additional annual federal subsidy outlays attributable to the warming would be approximately \$5.6 billion under the current practice mix, of which non-irrigated corn alone accounts for roughly \$4.4 billion. The program's projected exposure to warming is therefore concentrated almost entirely in non-irrigated production.

This paper makes two contributions to the literature on climate impacts and agricultural finance. First, it extends Diffenbaugh, Davenport, and Burke (2021) by switching from backward-looking attribution of historical warming to forward-looking projection under counterfactual scenarios, applied to a policy-relevant question: not how much past fiscal exposure came from warming, but where the future fiscal exposure is concentrated. Second, where existing projection work aggregates across crops, programs, and practices, we disaggregate to the level at which the FCIP rates policies. This decomposition shows that the program's projected warming exposure is concentrated in non-irrigated production, a pattern that aggregate projections cannot reveal. We read

this concentration as descriptive of the existing stock of irrigated acres. It is not evidence that irrigation causes lower fiscal exposure, and it is not an argument for expanding subsidized irrigation, whose long-run viability depends on water availability and resource costs that lie outside our data.

The remainder of the paper is organized as follows. Section 2 develops the empirical framework. Section 3 describes the data. Section 4 presents the main regression results. Section 5 reports robustness checks. Section 6 develops the warming projections and quantifies the differential fiscal exposure of the program. Section 7 concludes.

2. Empirical Framework

Federal crop insurance pays a per-acre indemnity to producers when realized outcomes fall below a contractual guarantee. We focus on Revenue Protection (RP), the most widely used product in the program.

Under RP, per-acre indemnity, liability, and the loss cost ratio are:

$$\text{Ind} = \max\left\{0,\ \text{APH} \cdot \text{Cov} \cdot \max(P^{\text{proj}}, P^{\text{harv}}) \ - \ y \cdot P^{\text{harv}}\right\}, \tag{1}$$

$$\text{Liab} = \text{APH} \cdot \text{Cov} \cdot \max(P^{\text{proj}}, P^{\text{harv}}), \tag{2}$$

$$\text{LCR} = \frac{\text{Ind}}{\text{Liab}}, \tag{3}$$

where APH is the producer's Actual Production History yield, Cov is the elected coverage level (a fraction of APH, typically 65–85%), P^{proj} and P^{harv} are the projected and harvest prices, and y is realized yield. Both Ind and LCR are censored at zero. We use indemnity per acre and LCR as our two dependent variables.

Of the two, the loss cost ratio is the actuarially relevant object. Under the loss-cost approach that underlies federal crop insurance rating, the expected loss cost ratio equals the actuarially fair component of the premium rate, and expected indemnity per acre equals the actuarially fair premium per acre (Tack, Coble, and Barnett, 2018). Charged rates depart from this benchmark because they incorporate reserve and catastrophic loadings and are built from longer loss histories

than our sample period. We therefore read the regression coefficients as describing changes in the expected loss cost that map to the actuarially fair component of premiums and, through the subsidy schedule, to federal outlays. We treat indemnity per acre as the more directly interpretable per-acre dollar loss and the loss cost ratio as the object that maps to program premium rates. We develop this translation explicitly under counterfactual climate scenarios in Section 6.

We estimate the following reduced-form specification separately for corn and soybeans, and for each $Y_{icpt} \in \{\text{Ind}_{icpt}, \text{LCR}_{icpt}\}$:

$$\begin{aligned} Y_{icpt} = {} & \beta_1 \,\text{EDD}_{it} + \beta_2 \,\text{WD}_{it} + \beta_3 \,\text{WD}_{it}^2 + \beta_4 \,\text{Cov}_c + \phi \,\text{Irr}_p \\ & + \gamma_1 \,(\text{Irr}_p \times \text{EDD}_{it}) + \gamma_2 \,(\text{Irr}_p \times \text{WD}_{it}) + \gamma_3 \,(\text{Irr}_p \times \text{WD}_{it}^2) \\ & + \gamma_4 \,(\text{Irr}_p \times \text{Cov}_c) + \alpha_{ip} + \delta_{tp} + \varepsilon_{icpt}, \end{aligned} \tag{4}$$

where i denotes county, c the coverage level, p the production practice, and t the year. EDD_{it} is cumulative growing-season Extreme Degree Days above 29°C (Schlenker and Roberts, 2009); WD_{it} is growing-season water deficit (mm), reference evapotranspiration minus precipitation, with WD_{it}^2 allowing nonlinearity; Cov_c is the elected coverage level; and Irr_p is a binary indicator equal to one for irrigated practices.

County-by-practice fixed effects α_{ip} absorb time-invariant heterogeneity at the county-practice level, including soils, long-run productivity, irrigation infrastructure, and persistent APH levels. The main effect of irrigation ϕ is absorbed by α_{ip} and is not separately identified. Year-by-practice fixed effects δ_{tp} absorb annual shocks common within practice. Identification of the weather coefficients relies on within county-practice variation in growing-season weather, with the identifying assumption that, conditional on the fixed effects, realized weather is orthogonal to county-level indemnities and loss cost ratios (Dell, Jones, and Olken, 2014; Blanc and Schlenker, 2017). Standard errors are clustered at the county level, and regressions are weighted by insured acres.

Because Y_{icpt} is either indemnity per acre or LCR, both bounded below at zero, a positive

coefficient corresponds to higher expected indemnities or a higher expected loss cost ratio. Positive coefficients are therefore associated with worse outcomes for both producers and the FCIP.

The marginal effect of EDD on the dependent variable equals β_1 for non-irrigated operations and $\beta_1 + \gamma_1$ for irrigated. The coefficient β_1 measures the change in expected indemnities (or LCR) per additional degree-day above 29°C for non-irrigated operations; γ_1 is the difference between irrigated and non-irrigated marginal effects.

The marginal effect of water deficit is nonlinear because of WD_{it}^2. For non-irrigated and irrigated operations respectively:

$$\left.\frac{\partial E[Y_{icpt}]}{\partial \mathrm{WD}_{it}}\right|_{\text{non-irr}} = \beta_2 + 2\beta_3 \mathrm{WD}_{it}, \tag{5}$$

$$\left.\frac{\partial E[Y_{icpt}]}{\partial \mathrm{WD}_{it}}\right|_{\text{irr}} = (\beta_2 + \gamma_2) + 2(\beta_3 + \gamma_3)\mathrm{WD}_{it}. \tag{6}$$

Because WD_{it} takes both negative values (excess moisture) and large positive values (drought-like conditions), the marginal effect changes sign across the support of WD_{it}.

We expect extreme heat to raise expected indemnities and loss cost ratios for non-irrigated operations ($\beta_1 > 0$). Given the agronomic evidence that supplemental water and transpirational cooling reduce heat damage, we expect a smaller response for irrigated operations ($\gamma_1 < 0$). For water deficit, we expect a U-shaped response, with losses elevated at both very negative WD (excess moisture) and very positive WD (drought-like conditions). This implies $\beta_2 < 0$ and $\beta_3 > 0$. On the drought side, we expect the irrigated response to be flatter ($\gamma_3 < 0$). We do not have a strong prior on γ_2, since supplemental water does not directly address excess-moisture losses.

Because irrigation status is not randomly assigned, the interaction coefficients $\gamma_1, \ldots, \gamma_4$ describe the differential association between weather and insured losses for operations classified as irrigated relative to those classified as non-irrigated within the same county and year. They do not identify the causal effect of applying irrigation water. Irrigated and non-irrigated systems differ along many dimensions beyond water application, including selection into irrigation infrastructure, soils and topography within counties, and contemporaneous management practices that travel with

irrigation status. The county-by-practice fixed effects absorb the time-invariant components of this heterogeneity, but cannot absorb time-varying differences in management or selection, and we have no instrument that isolates water application from these correlated factors. We therefore cannot separate the role of supplemental water from that of the other characteristics that distinguish the two production systems, and we interpret every irrigation differential in this paper as a descriptive association rather than a treatment effect.

3. Data

We combine county-level crop insurance outcomes from the USDA Risk Management Agency (RMA) with gridded weather data. The analysis covers 2011–2023 and focuses on corn and soybeans under Revenue Protection (RP). Observations are classified by irrigation practice using the RMA practice description, which separates irrigated and non-irrigated production within counties.

Insurance data come from the RMA Summary of Business. The first dependent variable, indemnity per acre, is the per-acre dollar payout reported in the Summary of Business. The second dependent variable, the loss cost ratio, is indemnity divided by liability as defined in Equation (3). Both outcomes are censored at zero by construction and capture the contract-defined loss region. We use the elected coverage level as reported by RMA, entered as Cov_c in Equation (4) together with its interaction with irrigation status.

Weather data are drawn from GridMET, which provides daily surface meteorological variables at 4-km resolution across the contiguous United States (Abatzoglou, 2013). We aggregate daily values to the county level using area-weighted averages and focus on the growing season (May 1 through September 30).

Extreme Degree Days (EDD) measure cumulative exposure above a damaging temperature threshold:

$$\text{EDD}_{ct} = \sum_{d} \max\left(\text{Tmax}_{cd} - 29°\text{C},\ 0\right). \tag{7}$$

The smooth temperature response in Section 5 relaxes this threshold and lets the data determine

where the damage function steepens.

Water deficit is defined as reference evapotranspiration minus precipitation, accumulated over the growing season:

$$\mathrm{WD}_{ct} = \mathrm{ET0}_{ct} - \mathrm{Precip}_{ct}. \tag{8}$$

Because this is an atmospheric measure, it does not incorporate supplemental irrigation water applied in irrigated systems. We include WD^2 to allow for nonlinearity consistent with excess moisture risk at low or negative WD and drought risk at high WD.

3.1 Summary Statistics

Table 1 presents summary statistics for the four samples used in this paper, separating irrigated and non-irrigated production within each crop. All statistics are weighted by insured acres.

Panel A describes the two dependent variables. Mean indemnity per acre is higher for non-irrigated than for irrigated operations in both crops ($47.87 vs $36.17 for corn, $20.64 vs $16.90 for soybeans). The same pattern holds for the loss cost ratio (0.091 vs 0.058 for corn, 0.060 vs 0.041 for soybeans). Standard deviations are large relative to the means in both samples, indicating substantial variation in realized losses across counties and years.

Panel B characterizes the weather variation that identifies our regression coefficients. Irrigated operations face higher mean EDD exposure than non-irrigated for both crops (343 versus 205 degree-days for corn, 354 versus 198 for soybeans). Mean water deficit is also higher for irrigated operations (434 mm versus 262 mm for corn, 339 mm versus 252 mm for soybeans). Standard deviations of both weather variables are large within each sample, confirming substantial within-practice variation in growing-season weather.

Panel C reports policy characteristics. Mean coverage levels are slightly lower for irrigated than for non-irrigated operations (0.721 vs 0.774 for corn, 0.709 vs 0.766 for soybeans). Mean insured acreage per county-coverage-year cell is roughly three to four times larger for non-irrigated than for irrigated operations in both crops, reflecting the larger geographic footprint of non-irrigated production in the FCIP portfolio.

Table 1: Summary Statistics

	Corn		Soybeans	
	Non-Irrigated	Irrigated	Non-Irrigated	Irrigated
Panel A: Outcomes				
Indemnity ($/ac)	47.87	36.17	20.64	16.90
	(81.99)	(66.32)	(32.36)	(34.72)
Loss Cost Ratio	0.091	0.058	0.060	0.041
	(0.162)	(0.109)	(0.101)	(0.082)
Panel B: Weather				
EDD (29°C, degree-days)	205.4	343.2	197.8	354.5
	(179.4)	(200.2)	(168.5)	(207.2)
Water Deficit (mm)	262.5	433.6	251.6	338.8
	(224.9)	(249.9)	(203.8)	(202.2)
Panel C: Policy				
Coverage Level	0.774	0.721	0.766	0.709
Insured Acres (mean)	6,481	2,334	6,500	1,733
Observations	125,510	56,701	119,709	44,947

Notes: Standard deviations in parentheses below the mean. Statistics weighted by insured acres. Sample is Revenue Protection policies, 2011–2023. Indemnity is the per-acre dollar payout. Loss Cost Ratio is indemnity divided by liability. EDD is cumulative growing-season degree days above 29°C (May 1 – Sep 30). Water Deficit is reference ET_0 minus precipitation accumulated over the growing season; negative values indicate precipitation exceeding evaporative demand. Coverage Level is the elected coverage as a fraction of APH. Insured Acres is the mean insured acres per county-coverage-year cell.

4. Results

Table 2 reports baseline estimates from Equation (4) for corn and soybeans under Revenue Protection. For each crop we present results for both dependent variables: indemnity per acre, in dollars,

and the loss cost ratio. The two columns within a crop tell the same qualitative story in different units. Indemnity per acre is the more directly interpretable per-acre dollar loss, while LCR is the actuarially relevant share of insured liability and the object that maps to program premium rates. We lead with the loss cost ratio when we summarize the irrigation differential, because it is the object that determines premium rates and the focus of the projections in Section 6.

For non-irrigated corn, an additional degree-day above 29°C is associated with $0.47 higher per-acre indemnities and a 0.0007 higher loss cost ratio (Table 2, Columns 1 and 2). A one-standard-deviation increase in EDD exposure (179 degree-days for non-irrigated corn; Table 1) is associated with approximately $84 higher per-acre indemnities and a 0.13 higher loss cost ratio. For non-irrigated soybeans, the per-degree-day associations are roughly an order of magnitude smaller ($0.12 per acre and 0.0003 in LCR; Columns 3 and 4), and a one-standard-deviation EDD shock (169 degree-days) is associated with approximately $20 higher soybean indemnities and a 0.06 higher loss cost ratio.

The interaction coefficients on Irrigated $\times$ EDD are large, negative, and statistically significant for both crops and both dependent variables, indicating a substantially smaller association between extreme heat and losses for operations classified as irrigated. The implied irrigated marginal effect of EDD, obtained by summing the EDD coefficient and its irrigated interaction, is approximately $0.03 per acre per EDD for corn indemnity (a 93% reduction relative to non-irrigated) and effectively zero for corn LCR (a 99% reduction, with the residual association statistically indistinguishable from zero). For soybeans, the corresponding values are approximately $0.03 per acre per EDD for indemnity (a 75% reduction) and 81% for LCR. This pattern is consistent with the agronomic role of supplemental water and the associated transpirational cooling on currently irrigated land (Carter et al., 2016; Tack, Barkley, and Hendricks, 2017), although for the reasons given in Section 2 we cannot attribute the smaller association to water application alone.

Table 2: Weather Effects on Indemnity and Loss Cost Ratio (Revenue Protection)

	Corn		Soybeans	
	Indemnity	LCR	Indemnity	LCR
	(1)	(2)	(3)	(4)
EDD (29°C)	0.4706^{***}	0.0007^{***}	0.1212^{***}	$3.39\times10^{-4***}$
	(0.0201)	(3.60×10^{-5})	(0.0074)	(2.44×10^{-5})
Irrigated × EDD	-0.4391^{***}	-0.0007^{***}	-0.0906^{***}	$-2.75\times10^{-4***}$
	(0.0265)	(4.31×10^{-5})	(0.0094)	(2.75×10^{-5})
Water Deficit	-0.1162^{***}	$-2.94\times10^{-4***}$	-0.0536^{***}	$-1.73\times10^{-4***}$
	(0.0076)	(1.91×10^{-5})	(0.0031)	(1.02×10^{-5})
Irrigated × Water Deficit	-0.0194	$1.02\times10^{-4***}$	-0.0133^{*}	3.11×10^{-5}
	(0.0137)	(2.61×10^{-5})	(0.0072)	(1.91×10^{-5})
Water Deficit^2 ($\times10^3$)	0.1321^{***}	0.0005^{***}	0.1230^{***}	$3.98\times10^{-4***}$
	(0.0177)	(4.47×10^{-5})	(0.0075)	(2.25×10^{-5})
Irrigated × Water Deficit^2 ($\times10^3$)	5.66×10^{-5}	$-3.42\times10^{-4***}$	-0.0606^{***}	$-2.62\times10^{-4***}$
	(0.0207)	(4.69×10^{-5})	(0.0099)	(2.70×10^{-5})
Fixed Effects	County × Practice, Year × Practice			
Observations	181,877	181,877	164,464	164,464

Notes: $^{*}p<0.10$; $^{**}p<0.05$; $^{***}p<0.01$. Standard errors clustered at the county level in parentheses; regressions weighted by insured acres. Columns labeled Indemnity have indemnity per acre (in dollars) as the dependent variable. Columns labeled LCR have the loss cost ratio (indemnity divided by liability) as the dependent variable. EDD is cumulative growing-season degree days above 29°C (May 1 – Sep 30). Water Deficit is reference evapotranspiration minus precipitation (mm), accumulated over the growing season. Coefficients on Water Deficit^2 and its interaction are multiplied by 10^3 for readability. All specifications include the elected coverage level and its interaction with irrigation as additional controls (estimates omitted for brevity); the main coverage effect is large, positive, and statistically significant in all four columns.

The water deficit coefficients show a nonlinear relationship between atmospheric moisture conditions and insured losses. Across all four columns, the linear WD term is negative and the quadratic term is positive, implying a U-shape, so predicted indemnities and loss cost ratios are lowest at intermediate water deficit and rise at both extremes. The implied turning point depends on the outcome. For the loss cost ratio it falls near typical conditions in both crops, at roughly 290

mm for corn and 220 mm for soybeans, against mean water deficit of 262 mm and 252 mm. For indemnity per acre the corn turning point is higher, near 440 mm, so a larger share of non-irrigated corn acres sits on the excess-moisture side of the minimum. Figure 1 traces the implied response curves relative to the sample mean WD. On the drought side, the right portion of each panel, the irrigated curves are clearly flatter than the non-irrigated curves for both crops and both dependent variables. The exception is indemnity per acre for corn (top-left panel), where the two curves have a similar shape, although losses under drought remain lower for irrigated operations. On the excess-moisture side, the very negative WD region at the left of each panel, the two curves have very similar shapes, indicating that both practices realize comparable increases in indemnities and loss cost ratios when WD turns sharply negative. This asymmetry is consistent with the biophysical role of supplemental water, which addresses evaporative demand and soil-moisture deficits but cannot mitigate excess precipitation.

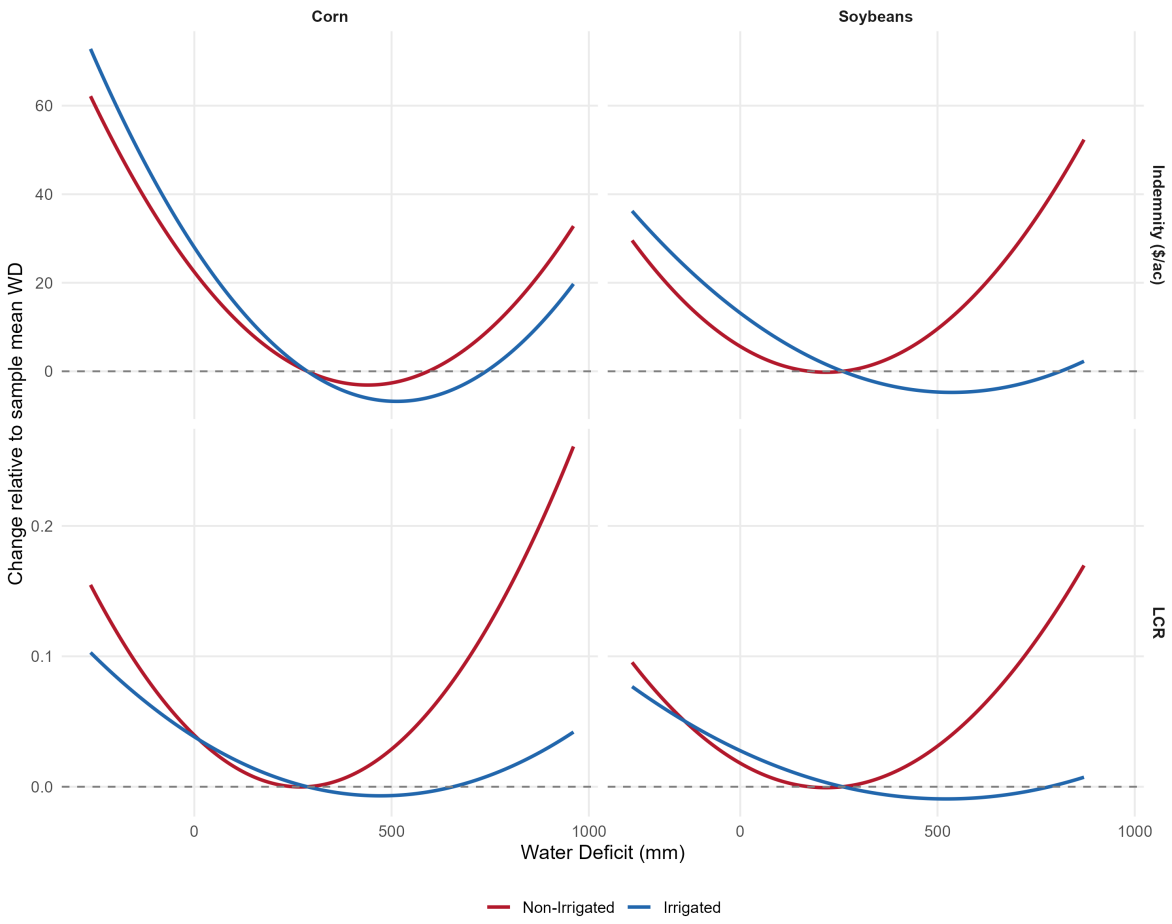


Figure 1: Water Deficit Effect on Indemnity per Acre and Loss Cost Ratio (Relative to Sample Mean WD)

Note: Change in predicted per-acre indemnity (top row) and loss cost ratio (bottom row) as water deficit varies relative to the sample mean, by crop (columns) and irrigation practice. Curves derived from Equation (4) using the coefficients in Table 2.

Overall, both heat and water deficit are associated with higher expected indemnities and loss cost ratios for non-irrigated production, while operations classified as irrigated show substantially smaller associations. Under loss-cost rating, these coefficients describe expected changes in the actuarially fair component of the premiums producers are charged and in the federal outlays paid as a share of those premiums. Section 6 develops this translation explicitly under counterfactual uniform-warming scenarios.

5. Robustness

We assess the sensitivity of the baseline estimates to three specification choices: the functional form of the temperature response, the level of geographic absorption in the fixed effects, and the potential for binning bias from trending temperature exposures. The first check reports results for both indemnity per acre and the loss cost ratio; the remaining two focus on indemnity per acre, with the loss cost ratio yielding quantitatively similar results up to scaling by liability.

5.1 Smooth Temperature Response

The baseline specification in Equation (4) imposes a piecewise linear damage function with a kink at 29°C. To assess whether this functional form drives our findings, we re-estimate the temperature response using a smooth fifth-order polynomial in daily growing-season temperature, following Schlenker and Roberts (2009). We aggregate daily maximum temperatures from GridMET into county-year exposures across 1°C bins from -5°C to 45°C using sine interpolation between daily $T_{\min}$ and $T_{\max}$, and we estimate the polynomial separately for irrigated and non-irrigated production within each crop, for both dependent variables. Water deficit and its square, the coverage control, and county and year fixed effects enter as in the baseline.

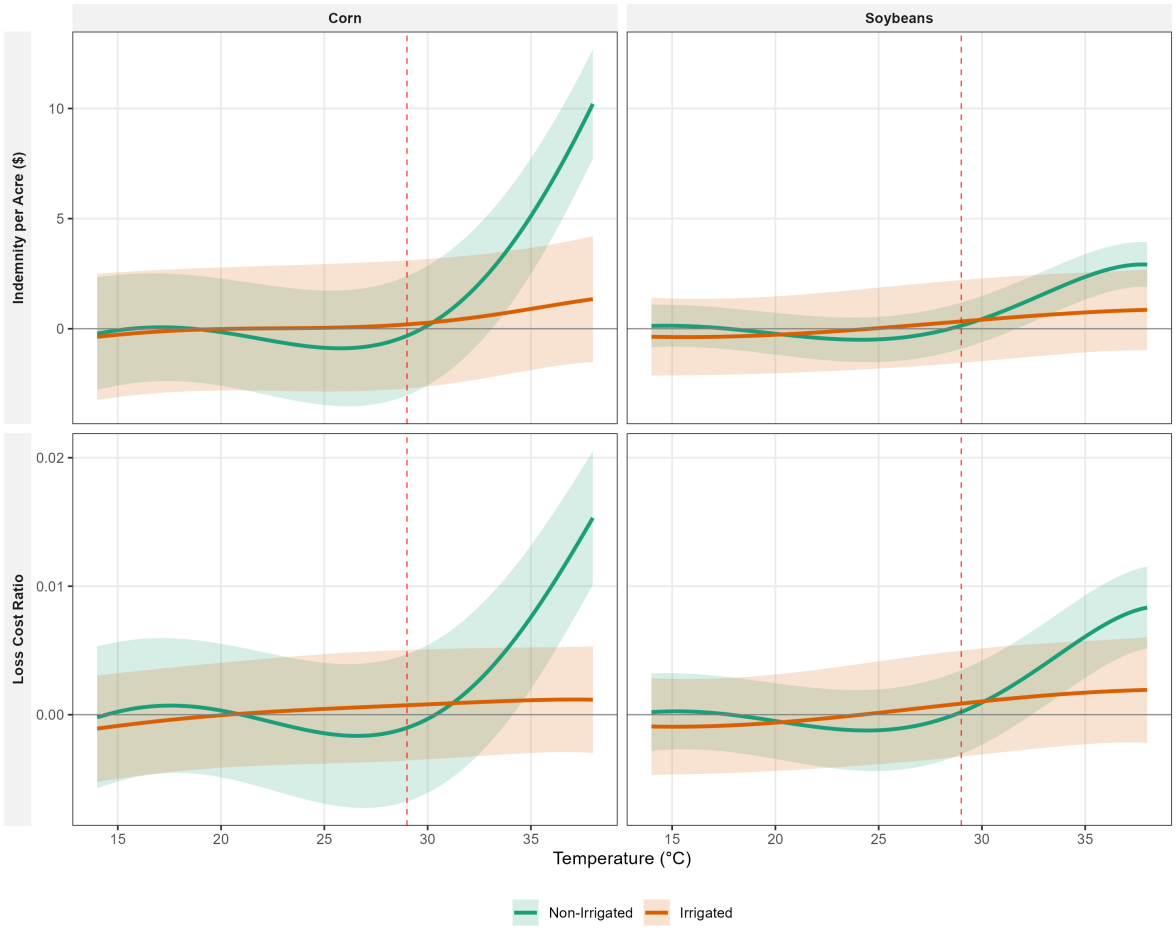


Figure 2: Smooth Temperature Response on Indemnity per Acre and Loss Cost Ratio

Note: Estimated effect of an additional day at temperature T on indemnity per acre (top row) and the loss cost ratio (bottom row), derived from a fifth-order polynomial in daily growing-season maximum temperature, estimated separately for each crop-practice combination. Curves are centered at the exposure-weighted sample mean. Shaded bands are 95% confidence intervals. The vertical dashed line marks the 29°C threshold used in the baseline EDD specification. Daily temperature exposures constructed using sine interpolation between $T_{\min}$ and $T_{\max}$ across 1°C bins from -5°C to 45°C.

Figure 2 traces the implied per-day effect on each dependent variable across the temperature range. For non-irrigated production, the response is essentially flat across the moderate range and rises above approximately 29°C for both crops and both outcomes, validating the threshold chosen in the baseline. For irrigated production, the response is nearly flat across the full temperature range and substantially smaller in magnitude than the non-irrigated response in every panel, confirming that the practice differential documented in Table 2 is not an artifact of EDD construction. The non-irrigated response also steepens with temperature above the threshold, a convexity we return

to when discussing the warming projections.

5.2 Alternative Fixed Effects

The baseline model absorbs county-specific production conditions with county × practice fixed effects and common annual shocks with year × practice fixed effects. We test two alternatives. The first replaces county × practice with state × practice fixed effects, so identification draws on both within-county and across-county-within-state variation in weather. The second adds state-year × practice fixed effects alongside the original county × practice fixed effects, absorbing state-level time-varying confounders, including regional price movements, state-level policy changes, and broad weather patterns, and identifying purely from cross-county variation within a state-year cell.

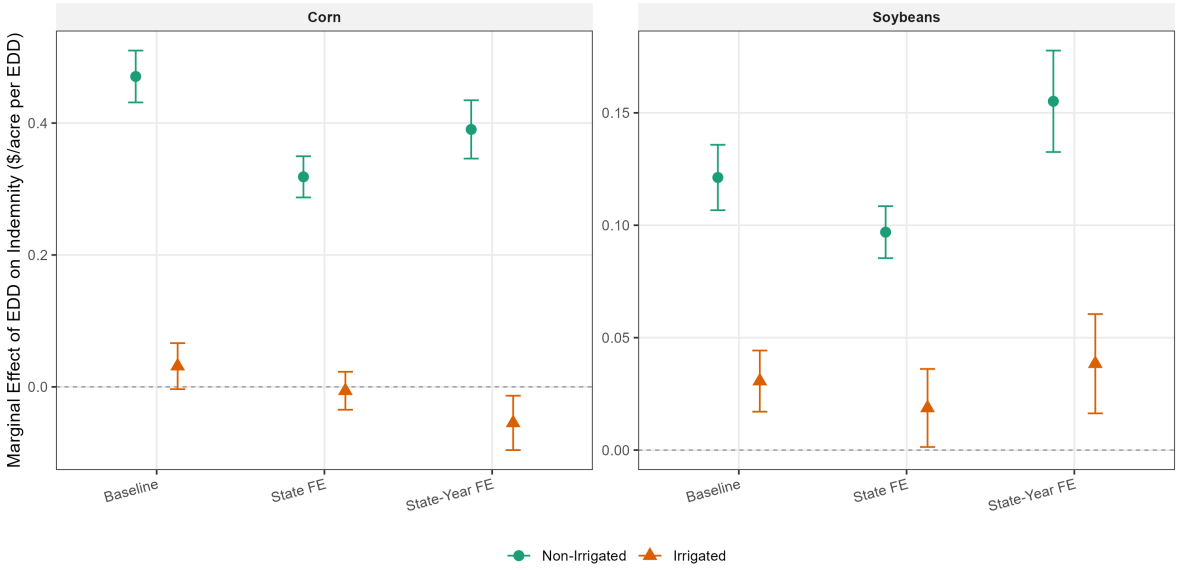


Figure 3: EDD Marginal Effects on Indemnity per Acre across Fixed Effects Specifications

Note: Point estimates and 95% confidence intervals for the marginal effect of EDD ($\geq$ 29°C) on indemnity per acre, by crop and irrigation practice. Baseline uses county × practice and year × practice fixed effects. State FE uses state × practice and year × practice fixed effects. State-Year FE uses state-year × practice and county × practice fixed effects. All specifications cluster standard errors at the county level and weight by insured acres.

Figure 3 compares the EDD marginal effects across the three specifications. For non-irrigated corn, the coefficient drops from 0.47 under the baseline to 0.32 under state × practice fixed effects, and partially recovers to 0.39 under state-year × practice fixed effects. For non-

irrigated soybeans, the pattern is qualitatively similar, although the state-year $\times$ practice specification yields a slightly larger coefficient (0.16) than the baseline (0.12). For irrigated production, the estimated marginal effect of EDD remains an order of magnitude smaller than the non-irrigated effect in every specification, ranging from a small negative value for corn under state-year fixed effects to small positive values for soybeans. Some of these irrigated estimates are statistically distinguishable from zero given the precision of the data, but all are economically negligible relative to the non-irrigated response. Across all four crop-practice combinations the qualitative conclusion matches the baseline: extreme heat raises per-acre indemnities for non-irrigated production, and the association is heavily attenuated for irrigated production.

Table 3: Binning Bias Correction: Baseline vs. Corrected (Indemnity per Acre)

	Corn (RP)		Soybeans (RP)	
	Baseline	Corrected	Baseline	Corrected
	(1)	(2)	(3)	(4)
EDD (29°C)	0.4706^{***}	0.4699^{***}	0.1212^{***}	0.1383^{***}
	(0.0201)	(0.0203)	(0.0074)	(0.0077)
Irrigated × EDD	-0.4391^{***}	-0.4673^{***}	-0.0906^{***}	-0.1051^{***}
	(0.0265)	(0.0248)	(0.0094)	(0.0098)
Water Deficit	-0.1162^{***}	-0.1176^{***}	-0.0536^{***}	-0.0564^{***}
	(0.0076)	(0.0077)	(0.0031)	(0.0031)
Irrigated × Water Deficit	-0.0194	-0.0105	-0.0133^{*}	-0.0067
	(0.0137)	(0.0131)	(0.0072)	(0.0072)
Water Deficit^2 ($\times 10^3$)	0.1321^{***}	0.1392^{***}	0.1230^{***}	0.1239^{***}
	(0.0177)	(0.0184)	(0.0075)	(0.0075)
Irrigated × Water Deficit^2 ($\times 10^3$)	5.66×10^{-5}	1.87×10^{-4}	-0.0606^{***}	-0.0650^{***}
	(0.0207)	(0.0215)	(0.0099)	(0.0099)
Expected EDD		2.3054^{***}		2.9608^{***}
		(0.4258)		(0.2255)
Irrigated × Expected EDD		0.1992		-1.7948^{***}
		(0.5938)		(0.2827)
Observations	181,877	181,877	164,464	164,464

Notes: $^{*}p < 0.10$; $^{**}p < 0.05$; $^{***}p < 0.01$. Standard errors clustered at the county level in parentheses. Regressions weighted by insured acres. The dependent variable is indemnity per acre. *Baseline* columns reproduce the main specification on the matched sample. *Corrected* columns add expected EDD and its interaction with irrigation as additional controls, following Jones et al. (2026). Expected EDD is constructed from county-month-specific warming trends in daily growing-season temperatures over 1979–2023. Coefficients on Water Deficit^2 and its interaction are multiplied by 10^3 for readability. All specifications include the coverage control and its interaction with irrigation (estimates omitted for brevity), and county × practice and year × practice fixed effects.

Table 3 reports the baseline and corrected estimates for both crops under Revenue Protection. The coefficient on expected EDD is large, positive, and statistically significant in both crops

(2.31 for corn, 2.96 for soybeans), indicating that the trending component of EDD is genuinely correlated with outcome trends in the panel. The correction is therefore doing real work, but its effect on the parameters of interest is modest. The EDD coefficient is essentially unchanged for corn (0.470 versus 0.471) and 14% larger for soybeans (0.138 versus 0.121). The interaction with irrigation strengthens slightly under the correction in both cases. The implied attenuation of the EDD association for irrigated production is 99% for corn and 76% for soybeans, broadly consistent with the baseline. The main finding, that extreme heat is associated with sharply higher per-acre indemnities for non-irrigated production and a much smaller response for irrigated production, is therefore not driven by binning bias from trending temperature exposure.

5.3 Bounded Outcomes, Coverage Control, and Spatial Correlation

Three further checks address the construction of the dependent variables, the coverage control, and the structure of the standard errors. First, both dependent variables are bounded below at zero, so we re-estimate the baseline using estimators that respect the bounded support: a fractional-response logit for the loss cost ratio and Poisson pseudo-maximum likelihood for both outcomes. Because coefficients are not comparable across link functions, we compare acre-weighted average marginal effects of EDD by practice. The implied attenuation of the EDD association for irrigated operations is essentially unchanged: for corn, 99% to 100% for the loss cost ratio and 93% to 95% for indemnity per acre across estimators; for soybeans, 74% to 81%. Second, the elected coverage level is chosen by producers and could respond to weather expectations, raising a bad-control concern. Dropping the coverage level and its interaction with irrigation leaves the EDD coefficients essentially unchanged for both crops and both outcomes. Third, county-level clustering allows arbitrary serial correlation within counties but not spatial correlation across counties. Clustering at the state level leaves all four non-irrigated EDD coefficients and their irrigated interactions statistically significant at the 1% level. Two-way clustering by county and year widens the intervals further, as expected with thirteen year clusters, but the EDD coefficients for non-irrigated production and their irrigated interactions remain statistically significant at the 5% level in both crops.

Point estimates are identical across clustering schemes by construction.

The robustness checks support the baseline findings. The temperature response is sharply nonlinear above 29°C for non-irrigated production and essentially flat for irrigated production regardless of how the response is parameterized. The practice differential is stable across alternative geographic absorption in the fixed effects, and the EDD coefficients are not materially affected by the binning bias correction of Jones et al. (2026).

6. Warming Scenarios: Premiums and Government Outlays

We use the regression coefficients reported in Table 2 to project how additional growing-season warming would translate into actuarially fair premiums, premium rates, and federal government outlays for the FCIP. We restrict attention to Revenue Protection policies for corn and soybeans, which together account for the largest share of FCIP insured acres and program subsidies. We apply uniform temperature shifts of $\Delta T \in \{1,2,3\}$°C to daily growing-season maximum temperatures. For each county, we recompute growing-season EDD under each warmed scenario, take the difference relative to the 2011–2023 baseline, and combine the resulting $\Delta\text{EDD}_{it}(\Delta T)$ with the estimated EDD coefficients to obtain projected changes in premiums and premium rates, separately for irrigated and non-irrigated practices. Under loss-cost rating, the actuarially fair component of the premium rate equals the expected loss cost ratio and the actuarially fair premium per acre equals expected indemnity per acre. The projected change in the actuarially fair premium per acre (Δp_{ipt}) under each warming scenario is then:

$$\Delta p_{ipt}(\Delta T) \;=\; \Delta E[\text{Ind}_{ipt}](\Delta T) \;=\; (\beta_1 + \gamma_1 \cdot \text{Irr}_p) \cdot \Delta\text{EDD}_{it}(\Delta T). \tag{9}$$

We hold water deficit at its baseline distribution in the main projections to isolate the temperature channel. Warming also raises reference evapotranspiration and therefore water deficit, a channel the heat-only projection omits. Because the estimated water deficit response is U-shaped, the sign of this omission is not uniform across counties: predicted losses fall with rising water

deficit on the excess-moisture side of the minimum and rise on the drought side. To quantify the omission, we construct an ET0-shifted scenario. We estimate the within-county sensitivity of growing-season ET0 to temperature from the daily GridMET data, approximately 7% per degree Celsius, shift each county's water deficit accordingly at each warming level holding precipitation fixed, and pass the change through the estimated water deficit coefficients. This empirical sensitivity exceeds the direct temperature term in Penman-Monteith calculations because warm years covary with higher radiation and vapor pressure deficit, so we read it as an upper bound on the evaporative-demand channel. The net effect on projected outlays is small relative to the temperature channel and differs in sign across crops: projected non-irrigated corn outlays fall modestly, because a majority of corn acres lie on the excess-moisture side of the indemnity minimum, while projected non-irrigated soybean outlays rise. The combined corn and soybean total moves by less than 3% in every scenario, reaching approximately $5.7 billion rather than $5.6 billion at +3+3 +3°C, and the concentration of projected exposure in non-irrigated production is unchanged. A separate consideration runs in the opposite direction: the per-day damage in Figure 2 accelerates above 29°C, while our projection applies a constant marginal effect per degree-day, so the linear extrapolation likely understates the non-irrigated response at the higher warming levels.

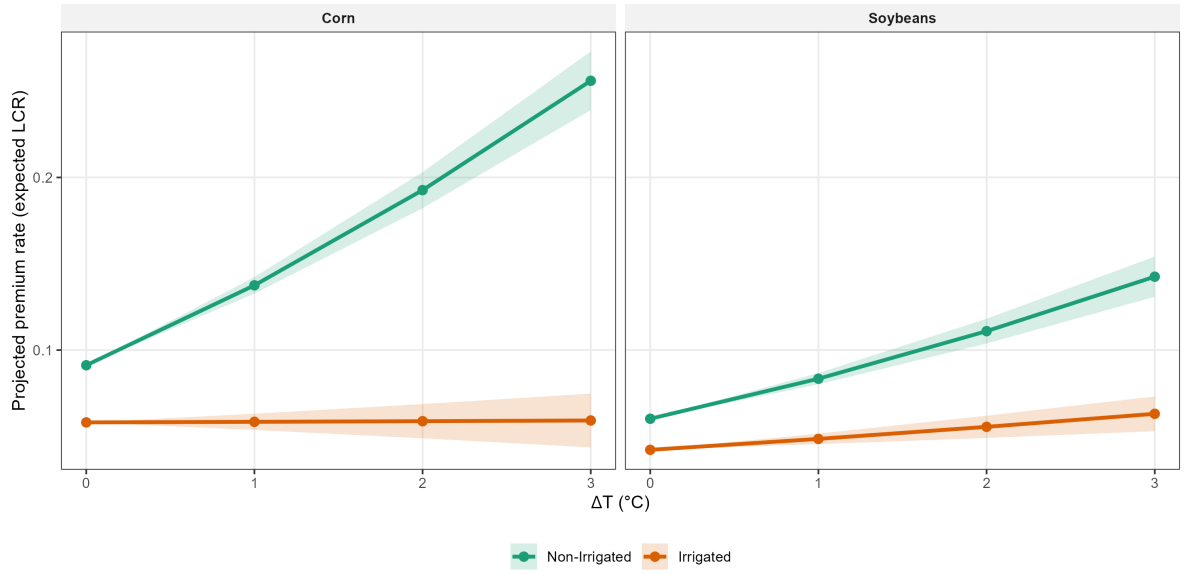


Figure 4: Projected Premium Rate under Uniform Warming, by Crop and Practice

Note: Projected actuarially fair premium rate (expected loss cost ratio) at $\Delta T \in \{0,1,2,3\}$°C of uniform warming, by crop (columns) and irrigation practice. Shaded bands show the uncertainty implied by the estimated EDD coefficients in Table 2, computed by the delta method. Projections apply uniform temperature shifts to daily GridMET maximum temperatures, recompute county-level EDD under each scenario, and translate into premium-rate changes using Equation (9). Water deficit is held at baseline.

Table 4 reports the projected premium per acre (Panel A) and premium rate (Panel B) at baseline and under each warming scenario. For non-irrigated production, both outcomes rise sharply with ΔT. Projected non-irrigated corn premium rises from \$48 to \$161 per acre at $+3$°C, with the corresponding premium rate rising from 0.091 to 0.256. The premium rate is the actuarially fair premium expressed per dollar of insured liability, so this implies that the fair rate for non-irrigated corn would nearly triple, from roughly 9 to 26 cents per dollar of liability. For non-irrigated soybeans, projected premium rises from \$21 to \$50 per acre, and premium rate from 0.060 to 0.143. For irrigated production, projected outcomes change very little across scenarios. Irrigated corn premium rises only from \$36 to \$46 per acre at $+3$°C, and irrigated corn premium rate stays essentially flat at 0.06.

Table 4: Projected Premium per Acre and Premium Rates under Uniform Warming

		Warming Scenario (°C)			
Crop	Practice	Baseline	+1	+2	+3
Panel A: Premium ($ per acre)					
Corn	Non-Irrigated	48	80	117	161
	Irrigated	36	39	42	46
Soybeans	Non-Irrigated	21	29	39	50
	Irrigated	17	20	24	27
Panel B: Premium Rate					
Corn	Non-Irrigated	0.091	0.138	0.193	0.256
	Irrigated	0.058	0.058	0.059	0.059
Soybeans	Non-Irrigated	0.060	0.083	0.111	0.143
	Irrigated	0.042	0.049	0.056	0.063

Notes: National acre-weighted averages of projected actuarially fair premium per acre (Panel A) and actuarially fair premium rate (Panel B), at $\Delta T \in \{0,1,2,3\}$°C of uniform warming, by crop and irrigation practice. Baseline ($\Delta T = 0$) corresponds to the acre-weighted mean over the 2011–2023 estimation sample. Under loss-cost rating, expected indemnity per acre equals the actuarially fair premium per acre, and the expected loss cost ratio equals the actuarially fair premium rate. Projections apply uniform temperature shifts to daily GridMET maximum temperatures, recompute county-level EDD under each scenario, and translate into premium changes using the EDD coefficients from Table 2. Water deficit is held at baseline.

To translate the per-acre premium changes into projected federal government outlays, we multiply the per-acre premium change (Δp_{ip}) by the practice-specific subsidy rate s_{ip} and by mean annual insured acres a_{ip}, then sum across counties:

$$\Delta\text{Outlay}_p(\Delta T) \;=\; \sum_i s_{ip} \cdot \Delta p_{ip}(\Delta T) \cdot a_{ip}. \tag{10}$$

Table 5 decomposes the projected change in annual federal government outlays into its three components. For each crop-practice combination at each warming scenario, column (1) reports the

acre-weighted subsidy rate, column (2) reports the projected change in actuarially fair premium per acre, and column (3) reports total annual insured acres. The final column reports the projected change in annual outlays, computed at the county level.

Table 5: Projected Change in Annual Federal Government Outlays under Uniform Warming

		(1)	(2)	(3)	(4)
Crop	Practice	Subsidy Rate (s)	Δp (\$/acre)	Acres (millions)	Δ Outlay (\$ millions)
Panel A: $\Delta T = +1°C$					
Corn	Non-Irrigated	0.62	31.8	62.6	1,241
	Irrigated	0.62	2.8	10.3	18
Soybeans	Non-Irrigated	0.63	8.3	59.9	314
	Irrigated	0.66	3.0	6.1	12
				Total at $+1°C$	**1,585**
Panel B: $\Delta T = +2°C$					
Corn	Non-Irrigated	0.62	69.6	62.6	2,717
	Irrigated	0.62	6.0	10.3	38
Soybeans	Non-Irrigated	0.63	18.1	59.9	686
	Irrigated	0.66	6.3	6.1	26
				Total at $+2°C$	**3,467**
Panel C: $\Delta T = +3°C$					
Corn	Non-Irrigated	0.62	113.0	62.6	4,410
	Irrigated	0.62	9.4	10.3	60
Soybeans	Non-Irrigated	0.63	29.4	59.9	1,112
	Irrigated	0.66	9.9	6.1	40
				Total at $+3°C$	**5,622**

Notes: Column (1) is the acre-weighted mean subsidy rate within each crop-practice group. Column (2) is the acre-weighted mean projected change in actuarially fair premium per acre at the indicated warming scenario, derived from Equation (9). Column (3) is total annual insured acres in millions. Column (4) is the projected change in annual federal government outlays in millions of dollars, computed at the county level as $\sum_i s_{ip} \cdot \Delta p_{ip}(\Delta T) \cdot a_{ip}$ per Equation (10); the column does not exactly equal the product of the row entries because subsidy rates, premium changes, and acres co-vary across counties within each crop-practice group. Water deficit is held at baseline.

Under uniform warming, projected additional annual federal government outlays for corn and soybeans rise from approximately \$1.6 billion at $+1°$C to approximately \$5.6 billion at $+3°$C.

Non-irrigated production accounts for the overwhelming majority of these projected outlays in every scenario, reflecting both its larger share of insured acres and its substantially higher EDD sensitivity (Table 2). At $+3$°C, non-irrigated corn alone accounts for approximately $4.4 billion in additional outlays, over three-quarters of the combined corn-and-soybean total. The program's projected exposure to warming is therefore concentrated almost entirely in non-irrigated production. These magnitudes are conditional on linearly projecting historical EDD sensitivities into warming scenarios beyond the observed range and on the current FCIP portfolio composition (insured acres, coverage levels, practice mix). The evaporative-demand channel, assessed above, alters them by less than 3%. We read them as a partial-equilibrium quantification of the temperature channel within the Revenue Protection corn and soybeans portfolio, and as a statement about where that exposure is concentrated, rather than as a full projection of FCIP costs.

7. Conclusions

This paper examines how extreme heat and water deficit relate to federal crop insurance indemnities, how that relationship differs between irrigated and non-irrigated production, and what the differential implies for federal government outlays under counterfactual warming. Using county-level administrative data from the USDA Risk Management Agency for 2011–2023 and daily weather from GridMET, we estimate how Extreme Degree Days above 29°C and growing-season water deficit relate to indemnities per acre and loss cost ratios under Revenue Protection, separately for irrigated and non-irrigated production. We then use the estimated weather sensitivities to project changes in actuarially fair premiums and program subsidy outlays under uniform-warming scenarios of $+1$, $+2$, and $+3$°C. Because irrigation status is not randomly assigned, we treat every irrigation differential as a descriptive association rather than the causal effect of applying water.

We find that operations classified as irrigated show a substantially smaller association between extreme heat and insured losses than non-irrigated operations. For corn under Revenue Protection, the implied heat sensitivity of the loss cost ratio is approximately 99% smaller under irrigation, and the residual association is statistically indistinguishable from zero. For soybeans, the

differential is approximately 81%. These results are robust to a smooth fifth-order polynomial in temperature following Schlenker and Roberts (2009), to alternative fixed effects specifications that absorb progressively more geographic and temporal variation, and to the binning bias correction of Jones et al. (2026) for trending temperature exposure. Water deficit follows a U-shaped relationship with insured losses, with both excess moisture and severe drought associated with higher per-acre indemnities. The two practices diverge on the drought side, where irrigated operations are less exposed, but respond similarly on the excess-moisture side.

Under uniform-warming scenarios, this differential weather sensitivity translates into a sharp concentration of projected federal government outlays in non-irrigated production. Projected additional annual outlays for corn and soybeans rise from approximately \$1.6 billion at $+1$°C to approximately \$5.6 billion at $+3$°C, with non-irrigated production accounting for the overwhelming majority in every scenario. At $+3$°C, non-irrigated corn alone accounts for approximately \$4.4 billion in additional annual outlays, more than three-quarters of the combined corn-and-soybean total.

These findings carry implications for the design and fiscal sustainability of the FCIP under continued warming. The program's projected exposure to warming is concentrated in non-irrigated production, the production that lacks the practices most often associated with buffering weather risk. Our results parallel those of Aglasan et al. (2024), who show that cover crop adoption is associated with lower crop insurance indemnities triggered by extreme heat, drought, and excess moisture. The pattern across studies is consistent with on-farm practices that buffer weather risk being associated with measurably lower insured losses, although neither study identifies a causal adaptation effect. We emphasize that the concentration we document is descriptive of the existing stock of irrigated acres. It is not an argument for expanding subsidized irrigation, whose long-run viability depends on water availability and resource costs, including groundwater depletion, that lie outside our data.

In terms of limitations, our analysis uses county-level aggregates and cannot capture within-county heterogeneity in irrigation infrastructure, water application rates, or management practices.

Because irrigation status is not randomly assigned, irrigated and non-irrigated operations differ along dimensions beyond water application, including soils, location, and management intensity. The practice differential we estimate should therefore be interpreted as a difference in realized weather sensitivity between these production systems rather than the causal effect of applying irrigation water alone. Our reduced-form estimates also do not separately identify the biophysical and behavioral components of the differential. Whether the smaller sensitivity of irrigated operations reflects transpirational cooling, maintained soil moisture, or correlated differences in management intensity is a question our data cannot resolve. Our uniform-warming projections also hold water deficit at its baseline distribution; because warming raises evaporative demand and would increase water deficit, the projections isolate the direct heat channel and, if anything, understate the non-irrigated exposure.

For future research, incorporating water availability constraints, including groundwater depletion trends and surface water allocation rules, would provide a more complete assessment of irrigation's long-run viability as an adaptation channel. Linking insurance-based estimates to farm-level yield data could help disentangle the biophysical and behavioral channels behind the irrigation differential.

References

Abatzoglou, J.T. 2013. "Development of Gridded Surface Meteorological Data for Ecological Applications and Modelling." *International Journal of Climatology* 33:121–131.

Aglasan, S., R.M. Rejesus, S. Hagen, and W. Salas. 2024. "Cover Crops, Crop Insurance Losses, and Resilience to Extreme Weather Events." *American Journal of Agricultural Economics* 106:1410–1434.

Blanc, E., and W. Schlenker. 2017. "The use of panel models in assessments of climate impacts on agriculture." *Review of Environmental Economics and Policy* 11:258–279.

Carter, E.K., J. Melkonian, S.J. Riha, and S.B. Shaw. 2016. "Separating Heat Stress from Moisture Stress: Analyzing Yield Response to High Temperature in Irrigated Maize." *Environmental Research Letters* 11:094012.

Dell, M., B.F. Jones, and B.A. Olken. 2014. "What Do We Learn from the Weather? The New Climate–Economy Literature." *Journal of Economic Literature* 52:740–798.

Diffenbaugh, N.S., F.V. Davenport, and M. Burke. 2021. "Historical Warming Has Increased U.S. Crop Insurance Losses." *Environmental Research Letters* 16:084025.

Jones, B.F., J. Moscona, B.A. Olken, and C. von Dessauer. 2026. "With or Without U? Binning Bias and the Causal Effects of Temperature Extremes." NBER Working Paper No. 34671, National Bureau of Economic Research, Cambridge, MA, Jan.

Schlenker, W., and M.J. Roberts. 2009. "Nonlinear Temperature Effects Indicate Severe Damages to U.S. Crop Yields Under Climate Change." *Proceedings of the National Academy of Sciences* 106:15594–15598.

Tack, J., A. Barkley, and N.P. Hendricks. 2017. "Irrigation Offsets Wheat Yield Reductions from Warming Temperatures." *Environmental Research Letters* 12:114027.

Tack, J., K. Coble, and B. Barnett. 2018. “Warming Temperatures Will Likely Induce Higher Premium Rates and Government Outlays for the U.S. Crop Insurance Program.” *Agricultural Economics* 49:635–647.